

EYA AZZABI

AI & Data Engineer | Final-Year Engineering Student

Ben Arous, Tunisia | +216 93 240 604 | eyaazzabi04@gmail.com | linkedin.com/in/eya-azzabi | github.com/EyaAzzabi | eyaazzabi.github.io

Available January–June 2027 for a 6-month final-year engineering internship | Open to relocation: Europe | Canada

PROFESSIONAL SUMMARY

Final-year Data Science engineering student at ESPRIT, completing a Master's in Data Science in parallel while working full-time as an AI & Backend Developer. I build **LLM agent systems, privacy-preserving ML pipelines and computer-vision models** for production.

PROFESSIONAL EXPERIENCE

AI & Backend Developer, Mafrah (formerly WebFullContact), Remote Sep 2025 – Present

- Parsing CVs meant sending personal data to a cloud LLM, creating GDPR exposure.* Engineered a **pseudonymisation-before-inference pipeline** (Presidio + spaCy NER → GPT-4o → local re-identification), so the model never receives candidate identity and **GDPR Art. 5(1)(c)** holds by construction.
- Keyword search could not match consultants to roles reliably.* Implemented **semantic matching** with vector embeddings in **PostgreSQL/pgvector**, replacing exact-term lookup with retrieval over meaning.
- Agents lost context between turns and services could not scale independently.* Built a **custom agent orchestration layer** with memory and deployed **containerised FastAPI microservices** behind a **NestJS** backend, now serving production users.

AI Consultant Intern, Talan Summer Camp, Talan Tunisie 2025

- Sales teams lost hours triaging inbound leads by hand.* Built **task-oriented LLM agents** for lead prioritisation, forecasting and automated outreach inside **Connect'IS**, delivering a working end-to-end platform with a cross-functional team within the two-month camp.

PROJECTS

JobReady: Master's thesis 2025 – 2026

Python, GPT-4o, pgvector, Presidio, spaCy, FastAPI, NestJS, Angular, Docker

- CV data is unstructured, and the EU AI Act classifies employment AI as high-risk.* Designed a three-phase system spanning information extraction, semantic retrieval and LLM-driven interview simulation, yielding a multi-tenant platform **GDPR-compliant by construction**. Defence autumn 2026.

QoS Buddy: Multi-agent telecom network diagnostics 2026

Python, scikit-learn, pandas, JavaScript; qos-buddy.vercel.app

- Operators detect SLA breaches only after they occur.* Developed **four autonomous agents** for latency prediction, anomaly detection, congestion analysis and data enrichment, reaching **$R^2 = 0.941$** on latency forecasting. ESPRIT integrated project, team of six.

Image Captioning with Visual Attention: Reimplementation and benchmark 2026

PyTorch, CNN encoders, cross-attention, Transformers

- Published captioning results are rarely independently reproduced.* Reimplemented *Show, Attend and Tell* (Xu et al., 2015) on Flickr8k against a Transformer decoder, achieving **BLEU-4 of 0.2224 versus the paper's 0.2130**; an entropy analysis confirmed attention concentrates on nouns over function words.

PCA of Tunisian Temperatures: Bachelor's thesis, extended and open-sourced 2024

Python, NumPy, Matplotlib; github.com/EyaAzzabi/pca-tunisia-temperatures

- Twelve monthly variables were highly correlated, with no geographic labels supplied.* Applied normed **PCA** across 16 stations, reducing them to **two components explaining 98.4%** of variance and recovering the coastal-inland divide unsupervised.

TECHNICAL SKILLS

Languages	Python, SQL, R, Java, TypeScript, JavaScript
Machine Learning	PyTorch, scikit-learn, XGBoost, TensorFlow, YOLO, CLIP, OpenCV, CNNs, Transformers, LoRA fine-tuning, feature engineering, model evaluation
NLP & LLMs	LangChain, RAG, prompt engineering, Hugging Face, spaCy, NLTK, TF-IDF, embeddings, pgvector, FAISS, vector databases
Data Engineering	pandas, NumPy, Apache Spark, Hadoop, ETL pipelines, PostgreSQL, MySQL, MongoDB, Elasticsearch, Snowflake
MLOps & Cloud	AWS, Docker, Kubernetes, CI/CD, GitHub Actions, model deployment, REST APIs, FastAPI, NestJS, microservices, Git, Linux, Agile/Scrum
Statistics	PCA, inferential statistics, predictive modelling, Power BI

EDUCATION

Engineering Degree in Data Science, ESPRIT, Tunis, Tunisia Sep 2025 – Oct 2027

Master's Degree in Data Science, University of Carthage, Bizerte, Tunisia Sep 2024 – 2026

Ranked **first in cohort**; taken concurrently with the ESPRIT engineering degree, self-funded.

BSc Mathematics and Computer Science, University of Carthage, Bizerte, Tunisia 2021 – 2024

LEADERSHIP & COMMUNITY

Speaker, WYSE (Women & Youth in STEM Employment) and Radio Amal, on women in technology; Soliya Connect 2025 – 2026
Program alumna

Project Developer, Enactus FSB; built SenseMate, an assistive AI device for blind users using real-time YOLO detection. 2023 – 2025
Won the **Early Stage award**, Enactus National Exposition 2024

Co-founder, Association Youth Clubs; founded the first youth club in Morneg, Ben Arous 2019 – 2021

CERTIFICATIONS & LANGUAGES

Applications of AI for Anomaly Detection, <i>NVIDIA</i>	Apr 2026	Developing Applications with LangChain, <i>DataCamp</i>	Feb 2025
Fundamentals of Deep Learning, <i>NVIDIA</i>	Jan 2026	Large Language Model Concepts, <i>DataCamp</i>	Feb 2025
AWS Academy Data Engineering, Graduate, <i>AWS</i>	May 2025	Getting Started with AI on Jetson Nano, <i>NVIDIA</i>	Jan 2025

Languages: Arabic (native) | French (advanced, C1) | English (advanced, C1) | German (basic, A2)